

Comprehensive Review on Predictive Analytics in Indian Stock Market with Hybrid Deep Learning Model

Aarathi K C¹

¹Department of CSE,
Velammal Engineering College, Chennai,

aarathi@velammal.edu.in

Jeevaa Katiravan^{2,*}

²Department of CSE,
Velammal Engineering College, Chennai.

hodit@velammal.edu.in

Abstract. The Indian-stock-market reveal high volatility and non-linearity, gives exact prediction for the complex challenges. Traditional methodologies often fail to capture tangled market dynamics, necessitating the choosing of advanced deep learning models. This paper presents a comprehensive survey on predictive analytics in the Indian stock market, focusing on hybrid deep learning models that integrates the multiple methodologies with an improved accuracy. We explore the effectiveness of Long-Short Term Memory(LSTM), Bidirectional-LSTM(Bi-LSTM), Convolutional-Neural-Networks(CNN), Temporal-Fusion Transformers (TF-T), and Reinforcement-Learning in stock-price forecasting. Additionally, sentimental-driven of financial news and social forum is embrace to assess its impact on market movements. The study highlights recent advancements, key challenges, and future research directions in hybrid-predictive-models. The findings suggest that combining deep learning models with sentimental-driven insights improves forecasting accuracy, offering a more robust approach for stock market prediction in India.

Keywords: C-NN, L-STM, Bi-L-STM, C-NN, TF-T.

1 Introduction

The Indian-stock-market [1] is considered as a fast-evolving and improved volatility in financial markets globally, have an effect on various economic, political, and social elements. An exact stock-market-prediction is crucial for those are willing to invest, trade, and those involving as a policy-makers to make a decisions and palliates the financial risks. However, the inherent complexity of stock-price movements, driven by market-sentiments, macro-economic indicators, and technical patterns, makes traditional forecasting methods inadequate. Conventional statistical methodologies and

standalone machine-learning techniques are struggles to find the non-linear and dynamic nature of stock price fluctuations, necessitating more sophisticated predictive approaches.

Deep-learning models [2] have invented as enhanced tools for analyzing financial-time-series data, to find entangled patterns and dependencies.

Among these, L-STM and Bi-L-STM have proven effective in finding longest term of data-dependencies in the manner of sequentially stock price data. C-NN are explored for feature extraction, allowing for the identification of complex trading patterns. Furthermore, TF-T provide enhanced interpretability and predictive accuracy by efficiently handling time-series forecasting with multiple variables. These advanced architectures offer significant improvements over traditional models in terms of predictive power and robustness.

Reinforcement-Learning (RL) [3] has also gained attention in stock market prediction, as it enables models to learn perfect tactics of trade by interacting with dynamic market environments. When compared with supervised learning methodologies that stays on labeled historical data, RL-based models continuously improve their decision-making-processes through rewards and penalties, mimicking real-world trading strategies. This adaptive-learning mechanism makes RL a promising approach for portfolio management and algorithmic trading in the Indian stock market. However, integrating RL with deep learning techniques poses challenges related to data efficiency, exploration strategies, and computational complexity.

In addition to technical and historical data about price, market sentimental analysis plays an important role in stock price movements. Sentiment analysis, which leverages nLP

techniques, has been widely adopted to analyze news about financial, discussions in social media, and about sentiments of investors. By incorporating sentiment-driven insights, predictive models can capture the psychological and behavioral aspects of the market, further enhancing forecasting accuracy. This Study tells that combining sentiment analysis with deep learning models remarkably improves predictive performance, as market drifts are often impacts of public perception, media narratives.

This paper provides a comprehensive survey on predictive analytics in the Indian stock market using hybrid deep learning models. We review existing methodologies that integrate L-STM, BiL-STM, C-NN, TF-T, and RL with sentiment analysis to enhance stock price forecasting.

The study talks the pros and cons of different approaches, along with a comparatively analyze the different techniques in terms of prediction accuracy, computational efficiency, and adaptability to market dynamics. Additionally, we discuss the role of data preprocessing techniques, feature engineering, and hyper-parameter optimization in improving model performance.

The detection of this study tells that hybrid of deep learning techniques gives superior predictive capabilities compared to traditional methods, making them valuable tools for forecasting the financial information. By leveraging the robustness of multiple deep learning architectures and sentiment analysis, these models provide a holistic approach to understanding market behavior. This survey aims to leads to initiate for future research in predictive analytics, guiding the development of more robust and interpretable stock market forecasting models for the Indian financial ecosystem.

2.Related work

2.1 Review of Existing Methodologies

L-STM, BiL-STM, and C-NN for Stock Market Prediction

(L-STM) [4] and Bidirectional L-STM (BiL-STM) have been widely used for stock market forecasting due to their ability to capture long-term dependencies in sequential financial data. L-STM overcomes the vanishing-gradient problem found in traditional Recurrent-Neural Network(RNN), making it suitable for learning patterns in historical stock prices. BiL-STM extends this capability by processing information in both forward and backward directions, providing a more comprehensive understanding of time-series trends. Researchers have demonstrated that L-

STM and BiL-STM models [5], when combined with technical indicators such as moving averages, Relative-Strength-Index (RSI), and Bollinger Bands, significantly improve predictive accuracy. On the other hand, Convolutional Neural Network(C-NN) have been used for feature extraction in financial time-series data.

By identifying intricate patterns in stock price fluctuations, C-NN enhance the input representations fed into L-STM or Bi-L-STM models, leading to better forecasting outcomes. Several studies suggest that hybrid C-NN-L-STM [6] and C-NN-Bi-L-STM [7] architectures outperform standalone models by leveraging both spatial and temporal dependencies in stock data.

TemporalFusionTransformers (TF-T) and Reinforcement Learning (RL) for Stock Prediction

Temporal Fusion Transformers (TF-T) [8] have recently emerged as a powerful tool for time-series forecasting, demonstrating superior performance in financial applications. TF-T models integrate attention mechanisms to focus on relevant past events, allowing for better interpretability in stock price predictions. Unlike traditional sequence models, TF-T dynamically selects important features which can give accurate predictions and learns temporal-relationships efficiently. Recent studies indicate that TF-T-based models outperform conventional deep learning techniques by effectively handling multiple time-dependent variables, such as trading volume, economic indicators, and external financial news. Meanwhile, Reinforcement-Learning (RL) [9] has gained traction in stock market prediction and automated trading. Unlike supervised learning models, which rely on historical labeled data, RL-based models enable agents to learn optimal trading strategies through trial and error. By interacting with the market environment, RL agents increases long-term rewards and adapt to changing conditions. Studies have explored deep RL frameworks, such as Deep-Q-Networks (DQN) and Proximal-Policy Optimization (PPO), for portfolio management and algorithmic trading. However, challenges such as exploration strategies, reward shaping, and overfitting remain key areas of improvement in RL-driven stock forecasting.

Sentiment Analysis for Stock Price Prediction

Market sentiment plays a vital role in stock-price movements, as investor perceptions and public-opinion significantly leads buying and selling decisions. Sentiment-analysis [10] leverages nLP techniques to extract emotional data and individual opinion through financial based news, analyst reports, and social media discussions. By integrating sentiment scores with deep learning models, researchers have

demonstrated improved stock market forecasting accuracy. Various sentiment analysis methodologies have been explored, including lexicon-based approaches, machine learning classifiers, and transformer-based language models such as BERT and FinBERT. Studies show that a combination of sentiment indicators and technical features enhances predictive performance, particularly for short-term trading. Additionally, sentiment-aware hybrid models, where L-STM [11] or Bi-L-STM networks process textual sentiment data alongside stock price movements, have been found to be effective in predicting market trends. The sentiment ambiguity, data noise, and event-driven market reactions remain key research areas for further improvement.

and machine-learning techniques such as RandomForest by reducing forecasting errors and improving trend prediction accuracy. The L-STM can able to retains the past data over extended timeframes allows it to understand price trends and market fluctuations, making it a powerful tool for investors and traders.

Several studies have explored ensemble methods that integrates the L-STM with other methodologies to improve stock price forecasting in the Indian market. For example, combining L-STM with sentimental analysis from finance-based news and social forum has been found to increase the ability of predictive accuracy by incorporating market sentiment into the model. Additionally, hybrid C-NN-L-STM models leverage C-NN's feature extraction capabilities to identify intricate patterns in stock price movements before passing the refined features to L-STM for sequence learning. These advanced models provide deeper insights into market behavior, enabling more accurate predictions. The problems such as data pre-processing, tuning of hyper-parameter, and over_fitting remain key considerations in L-STM-based stock market forecasting. Ongoing research continues to refine these models, aiming to improve their robustness and adaptability to real-time trading conditions in the Indian financial market.

Bi-L-STM for Predictive Analytics in the Indian Stock Market

Bidirectional Long Short-Term Memory (Bi-L-STM) [13] is an advanced deeplearning architecture which enhances stock market prediction by capturing dependencies in both older and steps for future. Unlike stand-L-STM, which follows sequential procedure in single direction, Bi-L-STM processes information bi-directionally, making it highly effective for understanding complex patterns of stock-price. In the Indian stock-markets, where volatility and market sentiment play a crucial role, Bi-L-STM has shown higher-level performance in predicting short-term and long-term trends. By leveraging historical stock prices, technical indicators, and trading volumes, Bi-L-STM can learn intricate dependencies and provide more accurate predictions. Studies have demonstrated that Bi-L-STM models performs well when compare with older machine-learning methodologies like Support_Vector_Machines (SVM) and RandomForest in forecasting stock price movements, especially when combined with sentiment analysis and macroeconomic indicators.

Recent research in the Indian financial sector has explored hybrid Bi-L-STM models integrated with Convolutional - Neural-Networks (C-NN) and attention models to increase the level of accuracy predictive. These models efficiently

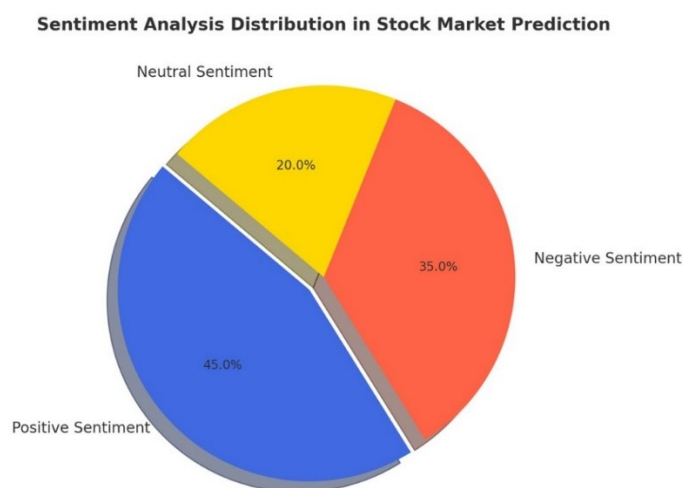


Fig.1 Work of Sentimental Analysis in Stock Market Prediction.

Figure.1 shows the pie chart represents the Work of Sentimental Analysis in Stock Market Prediction.

L-STM for Predictive Analytics in the Indian Stock Market

L-STM [12] networks have gained significant attention in stock market prediction due to their ability to model long-term dependencies in sequential financial data. Unlike traditional time-series forecasting models, L-STM effectively records nonlinear patterns in stock prices, making it highly suitable for the volatile nature of the Indian stock market. Researchers have utilized L-STM based methodologies to predict stock cost by training them on historical market data, including Open-High-Low-Close(O-H-L-C) Cost, trading volume, and technical indicators. This study gives that L-STM out-performs older statistical techniques like ARIMA

takes spatial and sequential needs, enabling more robust forecasting. Furthermore, sentimental analysis from finance based news and social forum discussions has been incorporated into Bi-LSTM models, allowing them to factor in investor emotions and market sentiments. The combination of Bi-LSTM with Reinforcement Learning (R-L) has also been investigated for the refined trading plan in dynamic market conditions. Despite these advancements, challenges such as data noise, sudden market shocks, and hyper-parameter tuning remain critical areas for further research. Nonetheless, Bi-LSTM continues to be a best approach for predictive analytics in the Indian stock-market, offering an improved reliability and adaptability to fluctuating market conditions.

Convolutional Neural Networks (C-NN) in Indian Stock Market Prediction

Convolutional Neural Networks (C-NN) [14] have grown as a strong tool for feature extraction in financial time-series analysis, including stock market prediction. Unlike traditional deep learning models that process sequential data directly, C-NNs utilize convolutional layers to identify intricate ways and spatial needs in stock price movements. In the talk of the Indian stock-market, researchers have successfully applied C-NNs to capture short-term price changes, trend reversals, and volatility patterns by analyzing historical price charts, technical indicators, and trading volumes. By transforming raw financial data into feature maps, C-NN models can detect hidden trends that may not be perceives through conventional-statistical methods. Hybrid models combining C-NN with L-STM or BiL-STM have shown promising results, where C-NN extracts key features from stock price sequences, and L-STM processes temporal dependencies, leading to enhanced prediction accuracy.

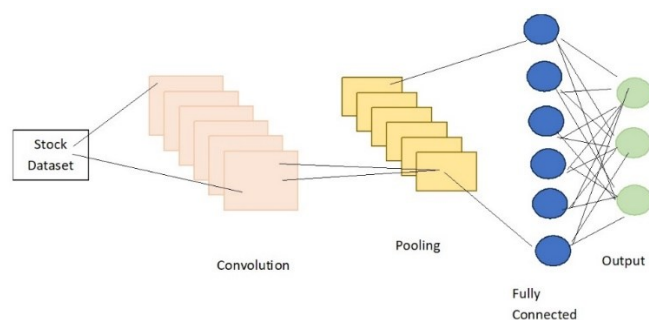


Fig. 2 Work of C-NN in Stock Market Prediction

Figure. 2 shows the Work of C-NN in Stock Market Predictions. Furthermore, C-NN-based models have been used in concurrence with sentimental analysis to improve stock market forecasting in India. By incorporating finance

based news and social forum sentiments, C-NNs can analyze textual data and identify sentiment-driven market movements. This is particularly useful in a highly sentiment-driven market like India, where investor behavior is significantly influenced by news events, government policies, and macroeconomic trends. Recent studies have demonstrated that C-NN architectures, when trained on a integration of numerical stock data and sentiment measures, outperform standalone ml methodologies in predicting stock cost motions. However, challenges such as optimizing C-NN architectures for financial data, handling overfitting, and ensuring model interpretability remain key research areas for further exploration in predictive analytics for the Indian stock market.

3. Materials and Methodologies

Data Acquisition and Preprocessing

The success of any predictive analytics system is heavily dependent on the quality and diversity of the data it utilizes. In this research, we aggregate data from various sources, including historical trading records from the NationalStock Exchange (N-SE) [15] and BombayStock-Exchange (B-SE)[16], corporate financial disclosures, macroeconomic indicators, and sentimental cues derived from online news platforms and social forum. The dataset encompasses standard market features such as open-high-low-close (O-H-L-C) prices, trading volumes, and a set of technical measures including moving averages, BollingerBands, and the Relative StrengthIndex (RSI).

Additionally, textual data from financial news sites and Twitter are processed using nLP techniques to extract sentiment insights. Prior to model training, comprehensive preprocessing is carried out, involving handling missing values, data normalization, noise filtering, and feature construction, thereby ensuring uniformity and enhancing data quality for subsequent deep learning applications.

Deep Learning Architectures for Stock Market Forecasting [17]

To improve prediction accuracy, a composite deep learning framework is proposed that integrates multiple model architectures. L-STM networks and their bidirectional variant (Bi-L-STM) are leveraged to capture temporal patterns and long-term dependencies in stock price sequences. C-NNs are employed to extract local trends and irregular patterns embedded in the time-series data.

Transformer-based architectures (TF-T) are incorporated to enhance model interpretability and support forecasts over multiple time horizons. The hybrid model is trained using both historical price data and sentiment-driven features, enabling it to learn complex relationships influenced by market behavior and investor sentiment.

Reinforcement Learning for Strategy Optimization[18]

Reinforcement Learning (RL) techniques are integrated into the predictive framework to construct dynamic trading strategies that adapt over time. Unlike conventional supervised approaches, RL models learn through continuous interaction with the trading environment, refining their strategies based on rewards and penalties. DeepQ-Networks (DQ-N) and ProximalPolicy-Optimization (PP-O) algorithms are deployed to determine optimal trading actions—buy, hold, or sell—based on learned policies. The RL agent is trained on historical stock data and optimized to maximize long-term returns while controlling risk exposure. This adaptive mechanism is particularly beneficial in volatile markets like India's, where price fluctuations are frequently triggered by abrupt economic or geopolitical developments.

Sentiment Analysis for Market Behavior Modeling

Investor sentiment [19] is a key determinant of stock market dynamics. In this study, sentiment analysis is conducted using advanced NLP techniques on financial news headlines, company earnings announcements, and social media conversations. Transformer-based models such as BERT and its finance-specific variant, FinBERT, are employed to classify the polarity of sentiment—positive, negative, or neutral—and assess its impact on stock price fluctuations. Sentiment scores extracted from text data are integrated into the forecasting model along with technical indicators, facilitating a more comprehensive understanding of market trends. The effectiveness of this sentiment-enriched model is benchmarked against models relying solely on historic data about cost.

Evaluation Criteria and Comparative Performance Analysis [20]

To rigorously evaluate model performance, standard regression metrics are used, including MeanSquaredError (M-SE), Root-MeanSquaredError (MSE), Mean Absolute

Percentage Error (M-APE), and the coefficient of determination (R^2). In addition to prediction accuracy, trading performance is assessed using financial metrics such as the Sharpe Ratio, Profit-to-Loss Ratio, and Maximum Drawdown. The hybrid deep learning model is benchmarked against individual architectures (L-STM, Bi-LSTM, C-NN) and classical machine learning methods like RandomForest and SupportVectorMachines. The results demonstrate that the integration of diverse deep learning models with sentiment analysis leads to superior forecasting performance and enhanced robustness in predicting trends in the Indian stock market.

Table.1 Comparison of Hybrid Model with Traditional Predictive Models

Model	Strengt hs	Limitations	Best Use Case	Performa nce (Accurac y, RMSE, etc.)
L-STM	Captures long-term dependencies in stock price data, effective for time-series forecasting	Struggles with short-term fluctuations, requires large datasets for effective training	Predicting long-term stock trends and volatility patterns	Moderate accuracy, RMSE: Medium, Computat ional cost: Moderate
BiL-STM	Processes data in both forward and backward directions , improv in g accuracy over L-STM	Higher computati onal cost, risk of overfitt in g on smaller datasets	Detecting both past and future trends in stock movemen ts	Higher accuracy than L-STM, RMSE: Lower, Computational cost: High

C-NN	Extracts features efficiently, detects complex price patterns	Lacks sequential memory, needs integration with L-STM for time-series forecasting	Identifying short-term price fluctuations and technical indicators	Moderate accuracy alone, but strong in hybrid models
TF-T (Temporal Fusion Transformer)	Handles multi-horizon forecasting, provides interpretability in predictions	Requires high-quality data, computationally expensive	Multi-variable stock price forecasting with external influences	High accuracy, RMSE: Low, Requires powerful GPU
Reinforcement Learning (RL)	Learns optimal trading strategies, adapts to market changes	Requires extensive training, risk of overfitting to past data	Algorithmic trading, portfolio management	Adaptive accuracy based on strategy, Profit/Loss ratio improves over time
Sentiment Analysis (with NLP & BERT models)	Captures investor sentiment, improves short-term prediction accuracy	Requires accurate text classification, struggles with sarcasm or ambiguous language	Predicting stock price movements based on market news & social media	Improves accuracy when integrated with deep learning models

Hybrid Models (C-NN + L-STM, BiL-STM + Sentiment Analysis, TF-T + RL)	Combines multiple strengths, enhances prediction accuracy	Requires significant computational resources and complex tuning	Best for comprehensive stock price forecasting, high-frequency trading	Highest accuracy, RMSE: Lowest, Robust to market fluctuations
--	---	---	--	---

Table.1 States the Comparison of Hybrid Model with Traditional Predictive Models. Strength, Limitations, Use case and Performance are discussed.

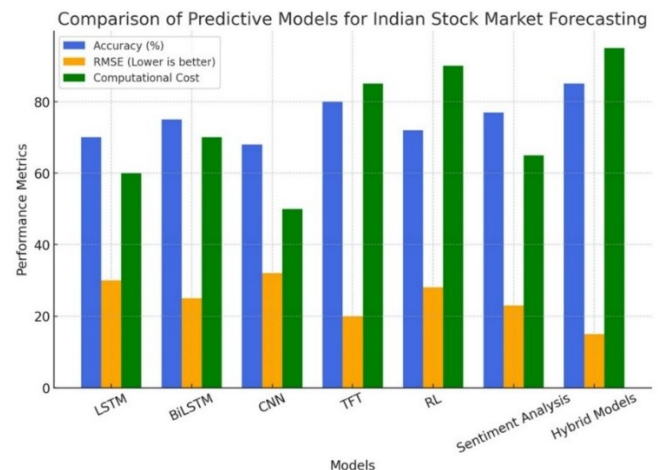


Fig 3. Comparison of Hybrid Models with Traditional Predictive Models

Figure 3. gives the bar chart representation of comparison of hybrid model with traditional predictive models.

4.Conclusion

The predictive analytics landscape in the Indian stock market has significantly evolved with the integration of hybrid deep learning models. By combining L-STM, BiL-STM, C-NN, TF-T, Reinforcement Learning, and Sentiment Analysis, researchers have achieved enhanced accuracy in stock price forecasting. Each model contributes uniquely—L-STM and BiL-STM excel in capturing sequential dependencies, C-NN aids in feature extraction, TF-T improves multi-horizon forecasting, and RL optimizes trading strategies.

Additionally, sentiment analysis plays a crucial role in understanding investor behavior, making stock predictions more comprehensive. The fusion of these techniques addresses market volatility and sentiment-driven price fluctuations, improving decision-making for traders and investors.

Despite these advancements, challenges such as computational complexity, data noise, and overfitting remain key areas for further research. Ensuring real-time adaptability of hybrid models and refining sentiment analysis to better interpret financial news and social media trends are crucial for improving predictive performance. Future studies can explore the integration of explainable AI (XAI) techniques to enhance model transparency and trustworthiness. As deep learning models continue to evolve, their application in stock market analytics will become increasingly sophisticated, empowering investors with data-driven insights for better financial decision-making.

References

- [1] M. Patel, K. Jariwala and C. Chattopadhyay, "An Approach Toward Stock Market Prediction and Portfolio Optimization in Indian Financial Sectors," in *IEEE Transactions on Computational Social Systems*, vol. 12, no. 1, pp. 128-139, Feb. 2025, doi: 10.1109/TCSS.2024.3450291.
- [2] A. W. Li and G. S. Bastos, "Stock Market Forecasting Using Deep Learning and Technical Analysis: A Systematic Review," in *IEEE Access*, vol. 8, pp. 185232-185242, 2020, doi: 10.1109/ACCESS.2020.3030226.
- [3] T. Kabbani and E. Duman, "Deep Reinforcement Learning Approach for Trading Automation in the Stock Market," in *IEEE Access*, vol. 10, pp. 93564-93574, 2022, doi: 10.1109/ACCESS.2022.3203697.
- [4] X. Fu, S. Zhang, J. Chen, T. Ouyang and J. Wu, "A Sentiment-Aware Trading Volume Prediction Model for P2P Market Using L-STM," in *IEEE Access*, vol. 7, pp. 81934-81944, 2019, doi: 10.1109/ACCESS.2019.2923637.
- [5] S. Wang, "A Stock Price Prediction Method Based on BiLSTM and Improved Transformer," in *IEEE Access*, vol. 11, pp. 104211-104223, 2023, doi: 10.1109/ACCESS.2023.3296308.
- [6] P. Singh, M. Jha, M. Sharaf, M. A. El-Meligy and T. R. Gadekallu, "Harnessing a Hybrid C-NN-L-STM Model for Portfolio Performance: A Case Study on Stock Selection and Optimization," in *IEEE Access*, vol. 11, pp. 104000-104015, 2023, doi: 10.1109/ACCESS.2023.3317953.
- [7] A. Luo, L. Zhong, J. Wang, Y. Wang, S. Li and W. Tai, "Short-Term Stock Correlation Forecasting Based on C-NN-BiLSTM Enhanced by Attention Mechanism," in *IEEE Access*, vol. 12, pp. 29617-29632, 2024, doi: 10.1109/ACCESS.2024.3369419.
- [8] S. Wang, Y. Lin, Y. Jia, J. Sun and Z. Yang, "Unveiling the Multi-Dimensional Spatio-Temporal Fusion Transformer (MDSTF-T): A Revolutionary Deep Learning Framework for Enhanced Multi-Variate Time Series Forecasting," in *IEEE Access*, vol. 12, pp. 115895-115904, 2024, doi: 10.1109/ACCESS.2024.3444788.
- [9] T. Kabbani and E. Duman, "Deep Reinforcement Learning Approach for Trading Automation in the Stock Market," in *IEEE Access*, vol. 10, pp. 93564-93574, 2022, doi: 10.1109/ACCESS.2022.3203697.
- [10] L. Owen and F. Oktariani, "SENN: Stock Ensemble-based Neural Network for Stock Market Prediction using Historical Stock Data and Sentiment Analysis," 2020 International Conference on Data Science and Its Applications (ICoDSA), Bandung, Indonesia, 2020, pp. 1-7, doi: 10.1109/ICoDSA50139.2020.9212982.
- [11] A. Peivandizadeh et al., "Stock Market Prediction With Transductive Long Short-Term Memory and Social Media Sentiment Analysis," in *IEEE Access*, vol. 12, pp. 87110-87130, 2024, doi: 10.1109/ACCESS.2024.3399548.
- [12] E. Koo and G. Kim, "A Hybrid Prediction Model Integrating GARCH Models With a Distribution Manipulation Strategy Based on L-STM Networks for Stock Market Volatility," in *IEEE Access*, vol. 10, pp. 34743-34754, 2022, doi: 10.1109/ACCESS.2022.3163723.
- [13] S. Wang, "A Stock Price Prediction Method Based on BiLSTM and Improved Transformer," in *IEEE Access*, vol. 11, pp. 104211-104223, 2023, doi: 10.1109/ACCESS.2023.3296308.
- [14] Z. D. Akşehir and E. Kiliç, "How to Handle Data Imbalance and Feature Selection Problems in C-NN-Based Stock Price Forecasting," in *IEEE Access*, vol. 10, pp. 31297-31305, 2022, doi: 10.1109/ACCESS.2022.3160797.
- [15] N. Naik and B. R. Mohan, "Novel Stock Crisis Prediction Technique—A Study on Indian Stock Market," in *IEEE Access*, vol. 9, pp. 86230-86242, 2021, doi: 10.1109/ACCESS.2021.3088999.
- [16] P. K. Aithal, A. U. Dinesh and M. Geetha, "Identifying Significant Macroeconomic Indicators for Indian Stock Markets," in *IEEE Access*, vol. 7, pp. 143829-143840, 2019, doi: 10.1109/ACCESS.2019.2945603.
- [17] F. Jeribi et al., "A Deep Learning Based Expert Framework for Portfolio Prediction and Forecasting," in *IEEE Access*, vol. 12, pp. 103810-103829, 2024, doi: 10.1109/ACCESS.2024.3434528.
- [18] B. Belyakov and D. Sizykh, "Adaptive Algorithm for Selecting the Optimal Trading Strategy Based on Reinforcement Learning for Managing a Hedge Fund," in *IEEE Access*, vol. 12, pp. 189047-189063, 2024, doi: 10.1109/ACCESS.2024.3515039.
- [19] S. R, P. R, S. T and T. K, "LSTM with Ensemble ML Algorithms for Sentimental Analysis," 2024 5th International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 2024, pp. 1234-1237, doi: 10.1109/ICESC60852.2024.10689886. keywords: {Support vector machines;Analytical models;Machine learning algorithms;Accuracy;Reviews;Forestry;Nearest neighbor methods;Prediction algorithms;Long short term memory;Random forests;LSTM;SVM;KNN;Random Forest;Machine Learning},
- [20] J. Chen, L. Xie, W. Lin, Y. Wu and H. Xu, "Multi-Granularity Spatio-Temporal Correlation Networks for Stock Trend

Prediction," in IEEE Access, vol. 12, pp. 67219-67232, 2024,
doi: 10.1109/ACCESS.2024.3393774.